

Aditya Shah

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Education

NYU Tandon School of Engineering

M.S. in Financial Engineering, CGPA: 3.95/4.00

Coursework: Stochastic Calculus, Derivative Securities, Time-Series Analysis, ML in Financial Engineering, Probability & Statistics, Quantitative Portfolio Management, Financial Risk Management, Algorithmic Trading, Commodities, Blockchain

New York, NY

Aug 2025 – May 2027

Nirma University

B.Tech. in Computer Science and Engineering, CGPA: 8.55/10

Ahmedabad, India

Aug 2021 – May 2025

Achievements

IAQF Student Competition 2026 – Winner (Team Captain): Led Team Sharpe Minds to win the 15th Annual International Association for Quantitative Finance competition among 31 teams from 15 academic programs, authoring “Pegged...Until It’s Not: Stablecoin Risk and Market Dislocation,” mapping findings to the GENIUS Act 2025

Experience

Traxys North America LLC

AI & Risk Management Intern

Jun 2026 – Aug 2026

New York, NY

- Co-engineered a Python VaR engine (Monte-Carlo, historical & parametric) with Risk Decomposition, P&L Drawdown & Stress Testing for a \$300M+ carbon-credits portfolio of inventory, forwards, options & ACP put-floor futures
- Built agentic workflows (Claude Code skills + routines) that automated 15+ daily price-load and risk-reporting pipelines across 10+ commodity desks for the Global Middle Office, cutting manual processing by 17+ hrs/week
- Monitored daily VaR utilization across 50+ trader groups, surfacing 10+ limit breaches monthly for timely risk action
- Automated daily analysis of forward curves and Greeks on cobalt derivatives to flag P&L swings >5%

Prowess Consulting LLC

Data Scientist Intern

Jan 2025 – May 2025

Ahmedabad, India

- Reduced 100K features to 200 principal components via Truncated PCA on 33K-ZIP-code Census data, cutting Snowflake query time by 60% and enabling K-Means clustering of 5 socio-economic profiles
- Launched QueryBot, a natural-language-to-SQL assistant (LangChain + GPT-4), cutting turnaround 2 hrs to 5 min

DRC Systems

Machine Learning Intern

Jun 2024 – Jul 2024

Gandhinagar, India

- Deployed BERT-based resume parsing (NER, QA, zero-shot) over 10,000+ resumes at 92% F1, saving 2+ hours daily

Teaching & Research

NYU Tandon School of Engineering

Teaching Assistant – Financial Risk Management

New York, NY

Sep 2026 – Present

- Supporting 20+ graduate students across market, credit, liquidity, operational and model risk, grading homework and quizzes and holding regular weekly office hours
- Guiding students through VaR, expected shortfall, PV01-neutral hedges, CDS-implied default rates and CVA

Teaching Assistant – Deep Learning Models in Finance

Mar 2026 – May 2026

- Fine-tuned SmolLM2 via LoRA-SFT and GRPO RL to call a Black-Scholes option-pricing tool, lifting tool-call parse rate from 0% to 100% and average reward from -0.117 to 1.033 with zero catastrophic forgetting
- Mentored 23 graduate students across two final projects, grading submissions and holding regular weekly office hours

Binghamton University

Research Assistant

Aug 2024 – Dec 2024

Remote

- Accepted at ICICT 2025: benchmarked 7 ML/DL models (LSTM, CNN, ANN, SVM, RF) on 20 years of equity data

Projects

Finance RAG Assistant: Built a RAG system (LangChain, FAISS, Streamlit, OpenAI) over 10+ SEC filings (10-K/10-Q), indexing 1,000+ chunks with top-3 retrieval, achieving 89% answer relevance and reducing API costs by 50%

Reasoning Model Pipeline (SFT + DAPO): Developed a 360M reasoning model (SmolLM2) via tokenizer surgery + LoRA-SFT on 1K DeepSeek-R1 traces, then DAPO RL with verifiable rewards, $2.25\times$ GSM8K accuracy

Market Anomaly Detection (Unsupervised ML): Built a point-in-time pipeline over 8 liquid ETFs ensembling Isolation Forest with rolling z-score detectors on return, volatility and volume features, capturing 41% of top-1% absolute-move days

Technical Skills

Languages & Databases: Python, SQL, R, C/C++, Java, PostgreSQL, SQL Server, Snowflake

ML/AI & Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face, PEFT/LoRA, TRL, LangChain, XGBoost, Random Forest, BERT, LSTM, CNNs, NLP, LLMs, RAG, RLHF (SFT, PPO, GRPO, DAPO), Transformers, Agentic AI, MCP, Regression

Tools: Git, CI/CD, AWS, FAISS, Pandas, NumPy, Streamlit, Power BI, Tableau, Excel, Bloomberg Terminal

Quant & Finance: Alpha Generation, Portfolio Optimization, Risk Modeling (VaR, CVaR), Factor Models, Options Pricing